

Video Tutor Agent

Paste a YouTube link, then **talk** to a tutor that answers **only from that video**, in English, Hindi, Hinglish or other Indian languages. Built entirely on **Sarvam**.

Code: github.com/tjay27/video-tutor-agent

Scope

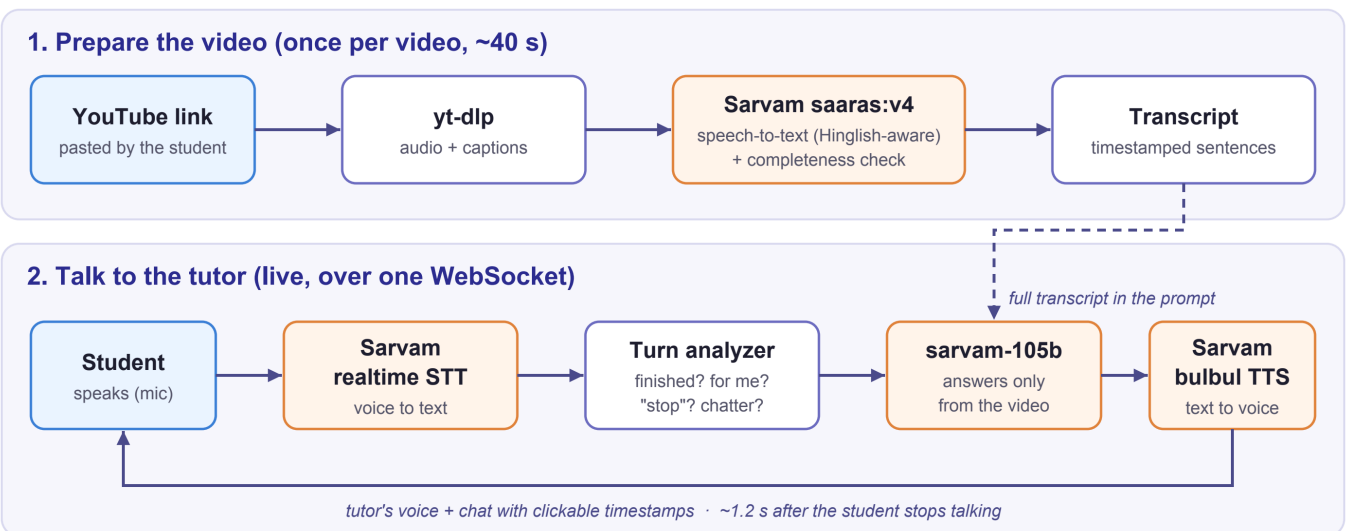
In scope

- Paste any YouTube link (up to 60 minutes) and the transcript is built automatically.
- Talk to the tutor by **voice, hands-free**. You can interrupt it at any time.
- Answers come **only from the video**, with timestamps you can click to jump there.
- Works in **English, Hindi, Hinglish** and other Indian languages.
- Copes with a real room: pausing to think, background chatter, saying "stop".

Out of scope

- Videos longer than 60 minutes (they would need audio splitting and RAG).
- Anything that is only shown on screen, such as slides, diagrams or code.
- Typed chat, user accounts, and saving history across sessions.
- Public hosting: it runs locally, because YouTube blocks downloads from cloud servers.

Architecture



Top 3 challenges

1. Knowing when the student has finished, and whether they are talking to the tutor

Problem: the tutor cut students off mid-thought, answered conversations with family, and froze while a song played.

Fix: a small, fast **turn analyzer** checks every utterance: is it finished, and is it meant for the tutor? It waits for unfinished thoughts and ignores chatter, TV and music.

2. Making interruptions feel natural

Problem: lowering the tutor's volume sounded odd, and speech got garbled when the student talked over the tutor on laptop speakers.

Fix: the tutor **pauses the moment the student speaks**. For a real question it stops and says "Sure." before answering; otherwise it continues from the same spot.

3. Accurate transcripts for Hinglish videos

Problem: YouTube's captions wrote English words in Hindi script, and **saaras:v3** silently skipped 2 minutes of one video.

Fix: **saaras:v4** in code-mix mode with the channel's key terms, plus a check against the captions that fills any gaps.